Two-Substrate Theory and the Genesis–Guidance–Transformation–Void Cycle

A formal framework for relations among Space, Matter, Energy, and Void

Kevin T.N.
jkdkr2439@gmail.com
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Abstract

This document compiles and formalizes Kevin T.N.'s *Two-Substrate Theory* and the *Genesis–Guidance–Transformation–Void* cycle. The framework proposes a unified view of relations among Space (as the substrate of matter), Time (as an intrinsic attribute of energy), and Void (as a third substrate containing potentiality and accounting for dissipation or "loss"). Concepts are defined explicitly, axioms are stated, and consequences are derived using differential, algebraic, and geometric forms. This anchor draft is intended as a starting point for simulation, falsifiable prediction design, and later refinement.

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1 Introduction: Epistemic Principles

Postulate 1 (Tri-partite reality (UTC))**.** We distinguish three layers of reality:

- **R1 (subjective):** personal experience, sensation, consciousness. Not directly formalizable.
- **R2 (intersubjective):** language, symbols, mathematics. Formalization occurs here.
- **R3 (objective):** reality independent of belief. Accessible only through traces.

Postulate 2 (Principle of formalization)**.** 1. A concept must be formed before it is assigned a symbol.

2. Every formalization is valid only within a specified frame of reference (denoted $[\mathcal{L}]$).
3. There is no "absolute truth" in this document—only domains where a formalism works, and domains where it does not.

Postulate 3 (Nature of mathematics)**.** Mathematics is not assumed to be a discovery of a metaphysical platonic realm. Instead, it is treated as a **language for describing structure**—a consequence of entities occupying space and interacting.

2 The Two Fundamental Substrates (and the Third Substrate)

2.1 Space: the substrate of matter

Definition 1 (Space)**.** Space $\mathcal{X}$ is modeled as a differentiable manifold equipped with a metric:

$$\mathcal{X} = (\mathcal{M}, g),$$

where $\mathcal{M}$ is a set of points and g determines distances.

Axiom 1 (Space is the substrate of matter)**.** Every material entity exists in $\mathcal{X}$ and occupies a definite location.

Definition 2 (Spatial point)**.** A point in space is denoted $\mathbf{x} \in \mathcal{X}$. In Euclidean 3-space:

$$\mathbf{x} = (x_1, x_2, x_3) \in \mathbb{R}^3.$$

2.2 Time: an attribute of energy

Definition 3 (Time)**.** Time t is a one-dimensional real parameter with order and irreversibility:

$$t \in \mathbb{R}, \qquad t_1 < t_2 \Rightarrow \text{"before/after"}.$$

Axiom 2 (Time belongs to energy)**.** Time is not treated as a substrate of matter; it is treated as an **intrinsic attribute of energy**. Energy exists and changes with time.

2.3 Void: the third substrate

Definition 4 (Void)**.** Void $\mathcal{V}$ is a space of potential states with no direct position in $\mathcal{X}$. It stores dissipation and traces of prior interactions. One possible realization is as a Hilbert space:

$$\mathcal{V} = \{\psi : \text{quantum states}\},$$

or more generally,

$$\mathcal{V} = \{\text{collections of possibilities}\}.$$

Axiom 3 (Void is a third substrate)**.** Void exists independently of space and time in the present formalism. Any dissipation in interactions is mapped into $\mathcal{V}$.

3 Primitive Entities

3.1 Matter

Definition 5 (Matter)**.** Matter is an entity occupying space, described by the set

$$\mathsf{Matter}(t) = \left\{ \left(\mathbf{x}_i(t), m_i, q_i, \mathbf{s}_i, r_i\right) \right\}_{i=1}^{N(t)},$$

where:

- $\mathbf{x}_i(t) \in \mathcal{X}$: position at time t
- $m_i \in \mathbb{R}^+$: mass
- $q_i \in \mathbb{R}$: electric charge
- $\mathbf{s}_i \in \mathbb{R}^3$: spin (intrinsic angular momentum, model-level variable)
- $r_i \in \mathbb{R}^+$: interaction radius (effective size)
- $N(t) \in \mathbb{N}$: number of material entities at time t

Axiom 4 (Impenetrability of matter)**.** Two material entities cannot occupy the same position at the same time:

$$\mathbf{x}_i(t) \neq \mathbf{x}_j(t) \qquad \forall\, i \neq j,\ \forall\, t.$$

3.2 Energy

Definition 6 (Energy)**.** Energy is modeled as a field acting on matter and producing change:

$$E : \mathcal{X} \times \mathbb{R} \to \mathbb{R}^k,$$

where $E(\mathbf{x}, t)$ denotes the energy-field value at $(\mathbf{x}, t)$.

Definition 7 (Energy decomposition)**.**

$$\begin{aligned} E_{\text{total}} &= E_{\text{bound}} + E_{\text{free}} + E_{\mathcal{V}}, \\ E_{\text{bound}} &= \sum_{i=1}^{N(t)} E_{\text{bound},i} \quad \text{(energy confined in matter)}, \\ E_{\text{free}} &= \int_{\mathcal{X}} \rho_E(\mathbf{x}, t)\, \mathrm{d}^3 x \quad \text{(free energy)}, \\ E_{\mathcal{V}} &= \text{energy stored in Void (potential)}. \end{aligned}$$

Axiom 5 (Conservation of total energy (including Void))**.** The total energy of the full system (including Void) is conserved:

$$\frac{\mathrm{d}}{\mathrm{d}t} E_{\text{total}} = 0.$$

3.3 Matter–energy relation and traces

Axiom 6 (Asynchrony)**.** Matter and energy belong to different reference systems and never fully synchronize. Interactions leave only traces.

Definition 8 (Trace)**.** A trace of an interaction is the set of observable changes in matter and energy after the interaction:

$$\begin{aligned} \Delta \mathbf{x}_i &= \mathbf{x}_i(t + \Delta t) - \mathbf{x}_i(t), \\ \Delta \mathbf{p}_i &= \mathbf{p}_i(t + \Delta t) - \mathbf{p}_i(t), \\ \Delta E_{\text{bound},i} &= E_{\text{bound},i}(t + \Delta t) - E_{\text{bound},i}(t), \\ \Delta \mathbf{s}_i &= \mathbf{s}_i(t + \Delta t) - \mathbf{s}_i(t). \end{aligned}$$

4 The Genesis–Guidance–Transformation–Void Cycle

4.1 Genesis

Definition 9 (Genesis)**.** Genesis is the appearance of new matter in space:

$$\mathsf{Matter}(t + \Delta t) = \mathsf{Matter}(t) \cup \{(\mathbf{x}_{\text{new}}, m, q, \mathbf{s}, r)\},$$

with $\mathbf{x}_{\text{new}} \in \mathcal{X}$ not occupied by other matter.

Axiom 7 (Conservation during Genesis)**.** Total energy is conserved during Genesis:

$$E_{\text{total}}(t + \Delta t) = E_{\text{total}}(t).$$

The energy required for Genesis may be drawn from E_{free} or $E_{\mathcal{V}}$.

4.2 Guidance

Definition 10 (Guidance)**.** Guidance is the process by which energy acts on matter and changes its state. The Guidance equations:

$$\frac{\mathrm{d}\mathbf{x}_i}{\mathrm{d}t} = \mathbf{v}_i(\mathbf{x}_i, E, t), \tag{1}$$

$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = \mathbf{F}_i(\mathbf{x}_i, E, \mathsf{Matter}, t), \tag{2}$$

where $\mathbf{p}_i = m_i\mathbf{v}_i$ is momentum and $\mathbf{F}_i$ is the net force on entity i.

Definition 11 (Forces induced by energy)**.** Forces can arise from a potential induced by the energy field:

$$\mathbf{F}_i^{(\text{field})} = -\nabla_{\mathbf{x}_i} U_i(\mathbf{x}_i, E),$$

and from direct interactions with other matter:

$$\mathbf{F}_i^{(\text{int})} = \sum_{j \neq i} \mathbf{F}_{ij}(\mathbf{x}_i, \mathbf{x}_j, E).$$

The total force may include stochastic environmental noise:

$$\mathbf{F}_i = \mathbf{F}_i^{(\text{field})} + \mathbf{F}_i^{(\text{int})} + \boldsymbol{\xi}_i(t).$$

4.3 Transformation

Definition 12 (Transformation)**.** Transformation is a state change in matter under energy action, comprising two simultaneous processes: Decay and Absorption.

Definition 13 (Decay)**.** Decay is the disappearance (removal) of one or more matter entities from space:

$$\mathsf{Matter}(t + \Delta t) = \mathsf{Matter}(t) \setminus \{\mathbf{x}_j\}_{j \in J},$$

together with release of energy:

$$E_{\text{free}} \leftarrow E_{\text{free}} + \sum_{j \in J} \Delta E_{\text{release},j},$$

with

$$\Delta E_{\text{release},j} = E_{\text{bound},j} + \sum_{k \neq j} \Delta E_{\text{interact},jk}.$$

Definition 14 (Absorption)**.** Absorption is the uptake of energy by matter, increasing its bound/internal energy:

$$E_{\text{bound},i} \leftarrow E_{\text{bound},i} + \sum_j \alpha_{ij} \Delta E_{ij},$$

where $\alpha_{ij} \in [0, 1]$ is an absorption coefficient dependent on interaction type and material properties.

Theorem 1 (Simultaneity of Decay and Absorption)**.** *In a Transformation event, Decay and Absorption occur simultaneously and are related through energy conservation:*

$$\sum \Delta E_{release} = \sum \Delta E_{absorb} + \Delta E_{\mathcal{V}},$$

where $\Delta E_{\mathcal{V}}$ is the portion transferred to Void.

4.4 Void (dynamic form)

Definition 15 (Void as a dynamic repository)**.** Void receives dissipated energy and stores information about the past:

$$\frac{\mathrm{d}E_{\mathcal{V}}}{\mathrm{d}t} = \sum_{\text{interactions}} \frac{\delta E_{\text{lost}}}{\delta t} - \sum_{\text{genesis}} \frac{\delta E_{\text{from } \mathcal{V}}}{\delta t},$$
$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = \text{information injected by interactions},$$

where $\mathcal{I}$ is an information field in Void.

Axiom 8 (No-loss property of Void (model axiom))**.** All information about the past is stored in Void as potential and can be re-instantiated when conditions are suitable.

5 Interactions

5.1 Interaction condition

Definition 16 (Interaction condition)**.** Two matter entities at $\mathbf{x}_i$ and $\mathbf{x}_j$ interact if

$$\|\mathbf{x}_i - \mathbf{x}_j\| \leq R_{ij},$$

where the interaction radius depends on properties and interaction type:

$$R_{ij} = f(r_i, r_j, \text{type}).$$

5.2 Matter–matter interaction types

Definition 17 (Replacement interaction)**.** One entity replaces another:

$$\mathbf{x}'_i = \mathbf{x}_j, \qquad \mathbf{x}'_j = \text{null}.$$

Bound energy update:

$$E'_{\text{bound},i} = E_{\text{bound},i} + \beta E_{\text{bound},j},$$

and loss to Void:

$$E_{\text{lost}} = (1-\beta) E_{\text{bound},j} \to \mathcal{V},$$

where $\beta \in [0,1]$ is a conversion coefficient.

Definition 18 (Destruction interaction)**.** Both entities disappear:

$$\mathsf{Matter} \leftarrow \mathsf{Matter} \setminus \{\mathbf{x}_i, \mathbf{x}_j\},$$

$$E_{\text{free}} \leftarrow E_{\text{free}} + E_{\text{bound},i} + E_{\text{bound},j}.$$

Definition 19 (Merge interaction)**.** Two entities merge into one:

$$\mathbf{x}_{\text{new}} = \frac{m_i \mathbf{x}_i + m_j \mathbf{x}_j}{m_i + m_j},$$

$$m_{\text{new}} = m_i + m_j - \Delta m,$$

$$E_{\text{bound,new}} = E_{\text{bound},i} + E_{\text{bound},j} - E_{\text{lost}},$$

$$\Delta m\, c^2 = E_{\text{lost}} \qquad \text{(if mass–energy conversion occurs).}$$

5.3 Energy interactions

Definition 20 (Energy absorption)**.** When incoming energy E_{in} acts on matter at $\mathbf{x}_i$, a fraction is absorbed:

$$\Delta E_{\text{bound},i} = \alpha_i(\mathbf{x}_i, E, t)\, E_{\text{in}},$$

where $0 \leq \alpha_i \leq 1$.

Definition 21 (Energy loss)**.** The non-absorbed portion is treated as loss to Void:

$$E_{\text{lost}} = (1 - \alpha_i) E_{\text{in}} \to \mathcal{V}.$$

Definition 22 (Energy propagation)**.** Free energy propagates in space via a diffusion-type equation:

$$\frac{\partial E}{\partial t} = \nabla \cdot (D \nabla E) + S(\mathsf{Matter}, E, t) - \gamma E + \eta(\mathbf{x}, t),$$

where D is a diffusion coefficient, S a source term, γ a dissipation rate, and η noise.

Definition 23 (Resonance)**.** If the energy frequency ω_E is close to the intrinsic frequency ω_i of matter, resonance increases absorption:

$$\alpha_i^{(\text{res})} = \alpha_i \cdot \eta(\omega_i, \omega_E),$$

with $\eta > 1$ for $|\omega_i - \omega_E| < \delta$, and $\eta \to 1$ otherwise.

6 Asynchrony and Reference-Frame Mismatch

6.1 Two substrate reference systems

Definition 24 (Spatial reference system)**.** Σ_X is the reference system for matter with characteristic quantities:

- Position: $\mathbf{x} \in \mathcal{X}$
- Distance: $d(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|$
- Volume: $V = \int \mathrm{d}^3 x$

Definition 25 (Temporal reference system)**.** Σ_T is the reference system for energy with characteristic quantities:

- Time: $t \in \mathbb{R}$

- Frequency: $\omega = \frac{\mathrm{d}\phi}{\mathrm{d}t}$
- Energy: $E = \hbar\omega$

6.2 Projection between systems

Definition 26 (Projection)**.** A projection $\Pi : \Sigma_X \times \Sigma_T \to \Sigma_{\text{common}}$ maps quantities from the two systems into a common frame to allow comparison.

Example 1 (Projection in special relativity)**.** In special relativity, the projection corresponds to the Lorentz transform:

$$\begin{cases} t' = \gamma\left(t - \dfrac{vx}{c^2}\right), \\ x' = \gamma(x - vt). \end{cases}$$

Example 2 (Projection in quantum mechanics)**.** In quantum mechanics, a projection is encoded via the Hamiltonian operator:

$$\hat{H}|\psi\rangle = i\hbar\,\frac{\partial}{\partial t}|\psi\rangle.$$

6.3 Mismatch point

Definition 27 (Mismatch point)**.** Let $A \in \Sigma_X$ and $B \in \Sigma_T$. After mapping to a common system, define the mismatch:

$$\Delta(A, B) = \|\Pi(A) - \Pi(B)\|.$$

Theorem 2 (Loss to Void (mismatch form))**.** *The dissipated portion in an interaction is the mismatch between input energy and absorbed energy after projection:*

$$E_{lost} = \Delta\big(E_{in},\, \Delta E_{bound}\big), \qquad \Delta E_{\mathcal{V}} = E_{lost}.$$

Definition 28 (Loss-detection tool)**.** The triple $(\Pi, \Sigma_{\text{common}}, \delta)$ is the tool for identifying lost energy:

- Π: projection between reference systems
- Σ_{common}: common frame
- $\delta(A, B) = \|\Pi(A) - \Pi(B)\|$: mismatch metric

7 Bound Energy and Mass

7.1 Energy confined by structure

Definition 29 (Bound energy)**.** Bound energy is the portion of energy confined within matter and not freely propagating. It is associated with eigenvalues of a wave equation under boundary conditions:

$$\nabla^2\psi + k^2\psi = 0 \quad \text{in region } \Omega_i,$$

$$\psi\big|_{\partial\Omega_i} = 0 \quad \text{(or periodic boundary conditions).}$$

One representation is:

$$E_{\text{bound},i} = \sum_n \hbar\omega_n = \sum_n \frac{\hbar^2 k_n^2}{2m_i}.$$

Theorem 3 (Mass as bound energy (model theorem))**.** *The mass of matter is proportional to bound energy:*

$$m_i = \frac{E_{bound,i}}{c^2},$$

where c is the speed of light in vacuum.

Corollary 1 (Higher density, higher bound-energy density)**.** *If $\rho_i = \frac{m_i}{V_i}$ is mass density, then bound energy per unit volume increases with ρ_i:*

$$\frac{E_{bound,i}}{V_i} = \rho_i c^2.$$

7.2 Spin as energy swirl

Definition 30 (Torque induced by energy)**.** Non-uniform energy action generates torque:

$$\boldsymbol{\tau}_i = \sum_{j\neq i} (\mathbf{x}_i - \mathbf{x}_j) \times \mathbf{F}_{ij} + \int_{\mathcal{X}} (\mathbf{x} - \mathbf{x}_i) \times \mathbf{f}_E(\mathbf{x})\, \mathrm{d}^3x,$$

where $\mathbf{f}_E(\mathbf{x})$ is the force density induced by the energy field.

Definition 31 (Spin)**.** Spin is an intrinsic angular momentum accumulated from torque over time:

$$\mathbf{s}_i = \int \boldsymbol{\tau}_i \,\mathrm{d}t + \mathbf{s}_i^{(0)}.$$

Theorem 4 (Quantization of spin (QM statement))**.** *In quantum mechanics, spin is quantized:*

$$\mathbf{s}_i^2 = \hbar^2 s(s+1), \quad s \in \left\{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\right\},$$

$$s_{i,z} = \hbar m_s, \quad m_s \in \{-s, -s+1, \dots, s\}.$$

8 Conservation Laws

Axiom 9 (Energy conservation)**.** Total energy of the full system is conserved:

$$E_{\text{total}} = E_{\text{bound}} + E_{\text{free}} + E_{\mathcal{V}} = \text{const}, \qquad \frac{\mathrm{d}}{\mathrm{d}t} E_{\text{total}} = 0.$$

Axiom 10 (Mass conservation (basic form))**.** In interactions without mass–energy conversion, total mass is conserved:

$$\sum_{i=1}^{N(t)} m_i = \text{const}.$$

Axiom 11 (Mass conservation (general form))**.** When mass–energy conversion is allowed:

$$\sum_{i=1}^{N(t)} m_i c^2 + E_{\text{bound}} + E_{\text{free}} + E_{\mathcal{V}} = \text{const},$$

equivalently the total-energy conservation statement.

Axiom 12 (Charge conservation)**.** Total electric charge is conserved in all interactions:

$$\sum_{i=1}^{N(t)} q_i = \text{const.}$$

Axiom 13 (Spin conservation)**.** In a closed system with no external torque, total spin is conserved:

$$\sum_{i=1}^{N(t)} \mathbf{s}_i = \text{const.}$$

9 Full Dynamical System

9.1 Matter evolution

$$\frac{\mathrm{d}\mathbf{x}_i}{\mathrm{d}t} = \frac{\mathbf{p}_i}{m_i}, \tag{3}$$

$$\frac{\mathrm{d}\mathbf{p}_i}{\mathrm{d}t} = \mathbf{F}_i(\mathbf{x}_i, E, \mathsf{Matter}, t) + \boldsymbol{\xi}_i(t), \tag{4}$$

$$\frac{\mathrm{d}\mathbf{s}_i}{\mathrm{d}t} = \boldsymbol{\tau}_i(\mathbf{x}_i, E, \mathsf{Matter}, t) + \boldsymbol{\zeta}_i(t), \tag{5}$$

where $\boldsymbol{\xi}_i, \boldsymbol{\zeta}_i$ are stochastic noises.

9.2 Energy evolution

$$\frac{\partial E}{\partial t} = \nabla \cdot (D \nabla E) + S(\mathsf{Matter}, E, t) - \gamma E + \eta(\mathbf{x}, t). \tag{6}$$

9.3 Void evolution

$$\frac{\mathrm{d}E_{\mathcal{V}}}{\mathrm{d}t} = \sum_{\text{interactions}} \frac{\delta E_{\text{lost}}}{\delta t} - \sum_{\text{genesis}} \frac{\delta E_{\text{from } \mathcal{V}}}{\delta t}, \tag{7}$$

$$\frac{\partial \mathcal{I}}{\partial t} = \Phi(\mathsf{Matter}, E, \mathcal{I}, t), \tag{8}$$

where $\mathcal{I}$ is a Void information field and Φ describes its update rule.

9.4 Boundary conditions

$$\begin{aligned}
&\mathbf{x}_i(t) \in \mathcal{X} \qquad \forall i, t \quad \text{(matter remains in space)},\\
&\lim_{\|\mathbf{x}\| \to \infty} E(\mathbf{x}, t) = 0 \quad \text{(energy decays at infinity)},\\
&E_{\mathcal{V}}(t) \geq 0 \qquad \forall t \quad \text{(Void energy nonnegative)}.
\end{aligned}$$

10 Consequences and Predictions (Anchor Statements)

Theorem 5 (Mass as an emergent phenomenon (model statement))**.** *Mass is not treated as a fundamental attribute of matter, but as a consequence of energy confined in material structure.*

Theorem 6 (Spin as a trace of energy swirl (model statement))**.** *Spin arises from asynchronous, non-uniform energy action on matter, producing accumulated torque; quantization reflects the structure of the relevant state space.*

Hypothesis 1 (Void as a potential repository (interpretive hypothesis))*.* Information about the past is stored in Void as potential and may be re-instantiated under suitable conditions. This hypothesis is a conceptual bridge to subjective phenomena (e.g., intuition, déjà vu) and should be treated as *non-verified* until linked to falsifiable predictions.

Theorem 7 (Asynchrony as a structural feature)**.** *Matter and energy never fully synchronize; every interaction leaves traces and a mismatch portion mapped into Void.*

Hypothesis 2 (Dark matter / dark energy (speculative corollary))*.* Dark matter and dark energy may be interpreted as (i) energy confined in structures not yet observed (dark matter), and (ii) Void energy participating in cosmic expansion (dark energy).

11 Conclusion and Extensions

11.1 Summary of the theory

This anchor draft formalizes the Two-Substrate Theory and the Genesis–Guidance–Transformation–Void cycle as a mathematical system featuring:

1. **Two primary substrates:** Space $\mathcal{X}$ (substrate of matter) and Time (attribute of energy), with **Void** $\mathcal{V}$ as a third substrate of potential and dissipation.
2. **Entities:** matter $\mathsf{Matter}(t)$ with position, mass, charge, spin; and energy $E(\mathbf{x}, t)$ as a field.
3. **Four phases:** Genesis, Guidance, Transformation (Decay + Absorption), and Void.
4. **Interaction types:** replacement, destruction, merge (matter–matter); absorption, loss, propagation, resonance (energy–matter).
5. **Conservation laws:** total energy (including Void), mass (appropriate form), charge, spin.
6. **Asynchrony:** matter and energy inhabit different reference systems; mismatch is mapped into Void.
7. **Bound energy:** mass as bound energy; spin as accumulated torque/swirl traces.

11.2 Development directions

1. **Numerical simulation:** implement an agent/particle simulation with $N \sim 10^3$ to 10^6, test conservation laws and derived behaviors.
2. **Empirical anchoring:** identify physical phenomena and design experiments to test mismatch-loss predictions and the role of Void.

3. **Mathematical extension:** model $\mathcal{V}$ as a symplectic manifold or structured state space; study topological invariants.
4. **Interdisciplinary mapping:** explore analogical applications in biology (cell cycles), sociology (social evolution), economics (business cycles), and epistemology (idea formation).

11.3 Closing statement

> *"Knowledge does not demand absolute proof; it demands knowing where it works and where it does not."*

This framework does not aim to replace modern physics. It aims to provide a **unified language** for describing systems from micro to macro scales, focusing on trace, mismatch, confinement, and potential.

A Appendix: List of Symbols

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Symbol	Meaning	Unit (if any)

Symbol	Meaning	Unit (if any)
$\mathcal{X}$	Space (substrate of matter)	–
$\mathcal{V}$	Void (third substrate of potential and dissipation)	–
$\mathsf{Matter}(t)$	Set of matter entities at time t	–
$\mathbf{x}_i$	Position of matter entity i	m
m_i	Mass of matter entity i	kg
q_i	Electric charge of matter entity i	C
$\mathbf{s}_i$	Spin (intrinsic angular momentum variable)	J·s
r_i	Interaction radius (effective size)	m
E	Energy field	$\mathrm{J/m^3}$ (model-dependent)
E_{total}	Total energy (including Void)	J
E_{bound}	Bound energy	J
E_{free}	Free energy	J
$E_{\mathcal{V}}$	Energy stored in Void	J
$\mathbf{F}_i$	Net force on matter i	N
$\boldsymbol{\tau}_i$	Torque on matter i	N·m
α_i	Absorption coefficient	–
β	Conversion coefficient in replacement interaction	–
η	Resonance gain factor	–
Π	Projection between reference systems	–
$\Delta(A, B)$	Mismatch between projected quantities A, B	(depends)
D	Energy diffusion coefficient	$\mathrm{m^2/s}$
γ	Dissipation rate	$\mathrm{s^{-1}}$
c	Speed of light in vacuum	m/s

$\hbar$ Reduced Planck constant J·s